

Final Research : Engineering Modeling

Study on airflow characteristics of rear wing of 2024 f1 car

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Abstract:

In Formula 1, the aerodynamic performance of a car's rear wing significantly influences track efficiency, particularly through its role in generating downforce and managing drag. This study focuses on two aerodynamic configurations of the rear wing with DRS (Drag Reduction System) open and closed, using 3D modeling in SolidWorks to simulate airflow in straight-line sections on the Monaco and Mexico circuits. At typical straight-line speeds on these circuits, computational analysis was conducted to examine the effects of DRS on aerodynamic loads, targeting optimal setups for both high downforce and low drag configurations. Results highlight the aerodynamic trade-offs associated with DRS activation, demonstrating reduced drag and increased efficiency in open mode, suitable for the high-speed straights at Mexico, while the closed configuration supports enhanced stability in Monaco's lower-speed sections.

Introduction:

Aerodynamics is one of the most important factors for 2024 Formula 1 cars. It influences not only the car's top speed but also its stability, handling, and cornering efficiency. The rear wing, is one of the parts that plays a role in generating downforce, which is the vertical force that helps the tires maintain better grip on the track, thus allowing higher cornering speeds. However, this downforce is often accompanied by an increase in drag, which can negatively affect straight-line speed. This creates a constant balancing act that F1 engineers must manage to achieve the best equilibrium between grip and speed performance.

In order to increase overtakes during races, the **Drag Reduction System (DRS)**, was introduced in 2011. This system allows drivers to open a section of the rear wing, reducing drag when they are in DRS zones, thus increasing straight-line speed. This function can only be activated under specific conditions, such as when a driver is within one second of the car ahead in the DRS zone defined by FIA.

In 2022 a new F1 regulations was introduced. This revision has focused on improving aerodynamic efficiency, with particular attention given to reducing drag.

Literature review:

1-FIA F1 regulation:

In the 2022 Formula 1 regulations, significant changes were made to the aerodynamic design, in order to improve the racing experience. One of the aspects of these changes was to reduce the drag. In this aim, the regulations focus on enhancing the ground effect of the floor. In consequence, the rear wing now contributing only about only 10% of the total downforce compared to previous designs. The main prepuce of this is reducing aerodynamic turbulence between cars.

Furthermore, these new rules set limits on the drag generated by the car. Specifically, the drag produced by the rear wing must be no more than 4% at a distance of 10 meters behind the car and 18% at 20 meters. This regulation aims to reduce the aerodynamic wake that affects trailing cars, allowing for closer racing and more overtaking opportunities. These changes have led to a redesign of the rear wing.

2-2022 Rear wing design:

Under the 2022 Formula 1 regulations, the rear wing dimensions are carefully controlled to ensure fair competition.

The rear wing has a **width of 1230 mm** and a **height of 600 mm**.

The flap angle, can reach a **maximum of 20°** in its normal configuration.

When the DRS is unable, the space between flap and wing should be **between 9.4mm and 13mm**.

When the DRS is activated, the flap opens by a maximum of **8.5 cm**.

Moreover, in the new F1 rear wing design, the flaps have a completely different shape compared to before. In fact, they don't really have an aerodynamic wing shape, but are more like a profiled board.



Figure 1: 2021 Mercedes Rear Wing.



Figure 2: 2022 McLaren Rear Wing.

3-Previews Rear Wing analysis:

To validate the accuracy of our model, we will rely on another study addressing the same subject. However, it is important to note that our study focuses on the **2022 regulations**, whereas all available studies are based on the **2016-2021 regulations**. Therefore, we will need to apply a coefficient to adapt the findings to our specific case.

The main differences between the two sets of regulations concern **downforce** and **turbulence**. In 2021, the rear wing generated approximately **25% of the car's total downforce**, compared to only **10%** in 2022.

Regarding turbulence, the 2021 regulations allowed for a drag level of **46% at 10 meters** behind the car and **35% at 20 meters**, which is far from the current limits of **18% at 10 meters** and **4% at 20 meters** under the 2022 regulations.

We will now review the results obtained in the study we are referencing.

First, we specify that all the results we will retain concern **rear wings with a 20° angle**, as this is the maximum angle allowed in our case.

In his study, **Zhidong Guo** focused on the variation of **downforce** and **drag** as a function of the **angle of attack** and the **speed** of the car. His findings showed that:

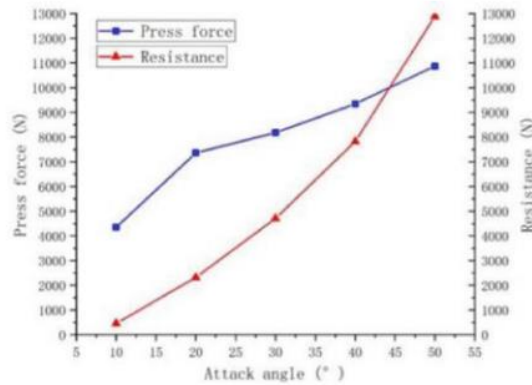


Figure 3: for a speed of 60m/s.

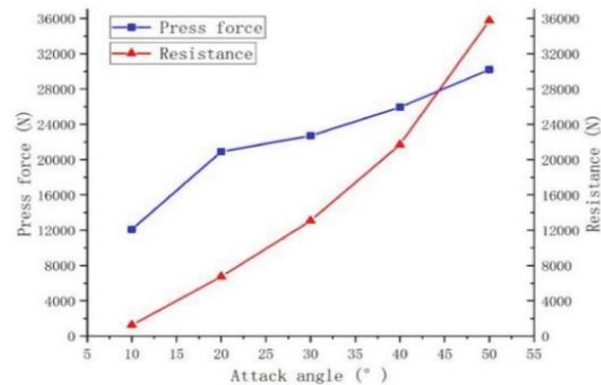


Figure 4: for a speed of 100m/s.

- At an angle of **20°**, the rear wing achieved the **optimal balance** between downforce and drag, with the best downforce-to-drag ratio observed.
- As the **speed increased**, both downforce and drag increased, but **drag** grew at a faster rate than downforce.

We will now adapt these results to align them with our **2022 regulations**.

Since the 2022 rear wing generates approximately **10% of the car's total downforce**, compared to **25%** in the previous regulations, we will apply a **correction factor** of 0.4 ($10\% \div 25\%$) to adjust the downforce values obtained in Zhidong Guo's study.

However, the total downforce in 2022 represents only 85% of the total downforce in 2021.

-For a speed of 60m/s: $7500 \times 0.34 = 2550$ N

-For a speed of 100m/s: $21000 \times 0.34 = 7170$ N

So for our research results we expected theses one.

For drag, given that the **2022 regulations** limit turbulence to **18% at 10 meters** and **4% at 20 meters**, a similar adjustment will be made to account for the stricter drag constraints compared to the **46% and 35% limits** in the previous regulations. This ensures that the adapted results reflect the reduced aerodynamic wake and overall drag levels mandated by the 2022 rules.

-For a speed of 60m/s: $2500 \times 0.34 = 850$ N

-For a speed of 100m/s: $7000 \times 0.34 = 2380$ N

So for our research results we expected theses one.

Methodology:

1-Flow Model:

1. Lift Equations

Lift generated by a Formula 1 rear wing depends on factors like the shape of the wing, the angle of attack, and the airspeed. Lift is created by the pressure difference between the top and bottom surfaces of the wing.

General Lift Formula:

Lift L is typically given by the following equation:

$$L = C_L \cdot \frac{1}{2} \cdot \rho \cdot V^2 \cdot A$$

Where:

- C_L is the **lift coefficient**, which is a function of the angle of attack, shape, and fluid state,
- ρ is the air density (approximately 1.225 kg/m^3 at ambient temperature),
- V is the airflow velocity,
- A is the surface area of the wing that experiences the pressure.

2. Turbulence Equations (Turbulence Model)

Turbulence is an important factor in the aerodynamics of vehicles, where air doesn't flow in a laminar manner. To simulate the effects of turbulence, we generally use **turbulence models**.

Turbulence Model:

In Flow Simulation, the most commonly used turbulence models are:

- **k-ε (k-epsilon)**: A two-equation turbulence model that is effective for simulating general incompressible turbulent flows. This model solves equations for turbulent energy (k) and turbulent dissipation (ϵ).

- **k- ω (k-omega):** Another turbulence model that is more accurate for simulations near walls (regions where flow is strongly affected by solid surfaces, like the wing).

Basic Turbulence Equations:

The k- ϵ and k- ω models are based on transport equations for turbulence:

1. Transport of turbulent energy (k):

$$\frac{\partial k}{\partial t} + \vec{u} \cdot \nabla k = P_k - \epsilon + \nabla \cdot (\mu_T \nabla k)$$

2. Transport of turbulent dissipation (ϵ for k- ϵ , ω for k- ω): In SolidWorks Flow Simulation:

$$\frac{\partial \epsilon}{\partial t} + \vec{u} \cdot \nabla \epsilon = C_{\epsilon 1} \frac{\epsilon}{k} P_k - C_{\epsilon 2} \epsilon^2 + \nabla \cdot (\mu_T \nabla \epsilon)$$

3. General Drag Formula:

$$D = C_D \cdot \frac{1}{2} \cdot \rho \cdot V^2 \cdot A$$

Where:

- C_D is the **drag coefficient**, which depends on the wing's shape, angle of attack, and the turbulent nature of the flow,
- ρ is the air density,
- V is the airflow velocity,
- A is the wing's surface area.

2-3D Model:

For the creation of our 3D model, we used SolidWorks 2023. Our model is based on the 2022 regulations, as well as the updates made between 2023 and 2024. Specifically, we used the rear wing from the 2024 McLaren MC38, which is used at Monaco and Mexico. Indeed, all Formula 1 teams have around 5

different front and rear wings depending on the specific requirements of each circuit.



Figure 5: Mc38 at Monaco.



Figure 6: Mc38 at Mexico.

The configurations of Monaco and Mexico are perfect for comparing with our reference study. Indeed, these are the circuits where aerodynamic downforce is at its maximum, making them the closest to the previous regulations.

Due to the complexity of this part, we had to create it entirely using surface modelling. Using the curve and thickening functions, followed by filleting, we were able to transition from our sketches to the final result.

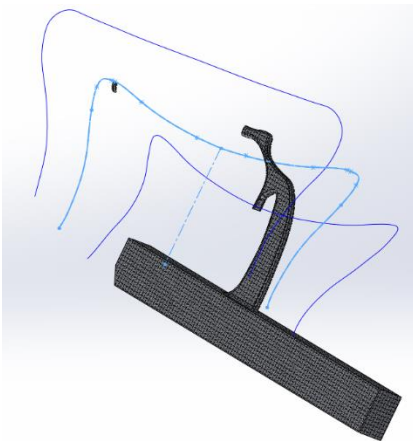


Figure 8: first step 3D model.



Figure 7: final main part.

Next, we duplicated it to create the DRS flap corresponding to our rear wing. Then, by adding the pivot connections, we obtained our main component.



Figure 9: DRS flap.

Finally, we modeled the parts that allow the rear wing to move, resulting in the final model.

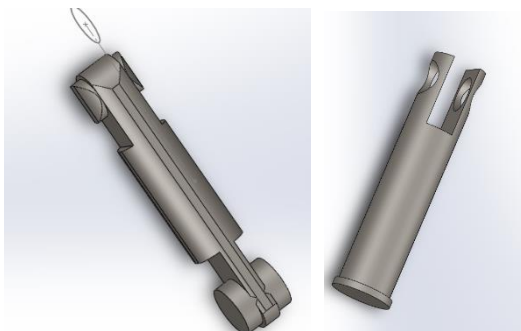


Figure 11: piston parts.



Figure 10: final 3D model.

Results and discussion:

Now, we have everything we need to move on to the simulation phase. We will therefore begin by describing our simulation environments.

1-Simulation environnements:

Monaco:

*During the 2024 Monaco Grand Prix, the maximum speed was achieved by Mercedes, reaching **289.9 km/h** or **80.528 m/s**. Additionally, the ambient temperature was **20.05°C**, with an atmospheric pressure of **1005 hPa**. The air density was **1.225 kg/m³**.*

Mexico:

During this edition of the Mexico Grand Prix, the maximum speed was set by

*McLaren, it was **362.5 km/h** or **99 m/s**. The ambient temperature reached **20°C**, and the atmospheric pressure was **1027 hPa**. The air density was **0.977 kg/m³** due to the **2300 m** altitude.*

Simulation setup:

For the simulation, I decided to use a mesh containing **400,000 cells** and **147 iterations** in order to get a precise understanding of the fluid behavior around the rear wing without requiring excessive computation time for the model.

Finally, we have **four flow studies** to perform:

- **Monaco** with and without DRS.
- **Mexico** with and without DRS.

2-Simulations:

Monaco :

Without DRS:

For our simulation under Monaco conditions, without opening DRS, we obtained the following results:

- Flow trajectorie:

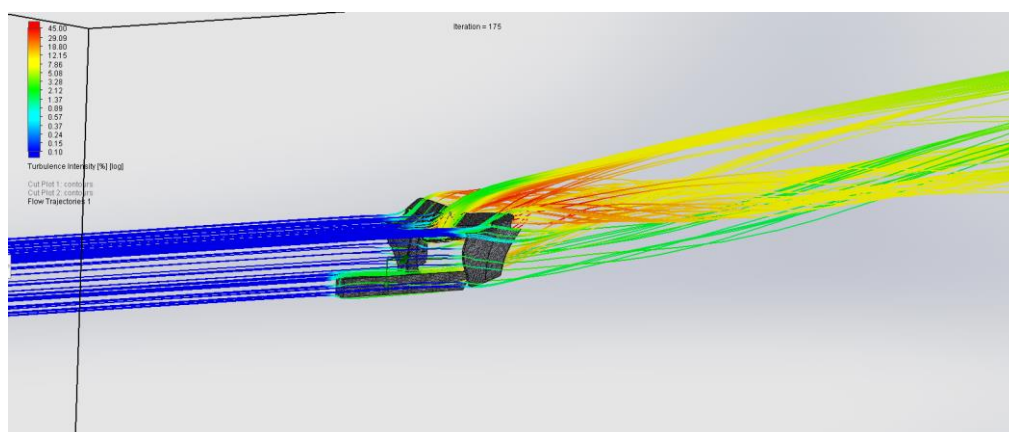


Figure 12: flow turbulence, front view.

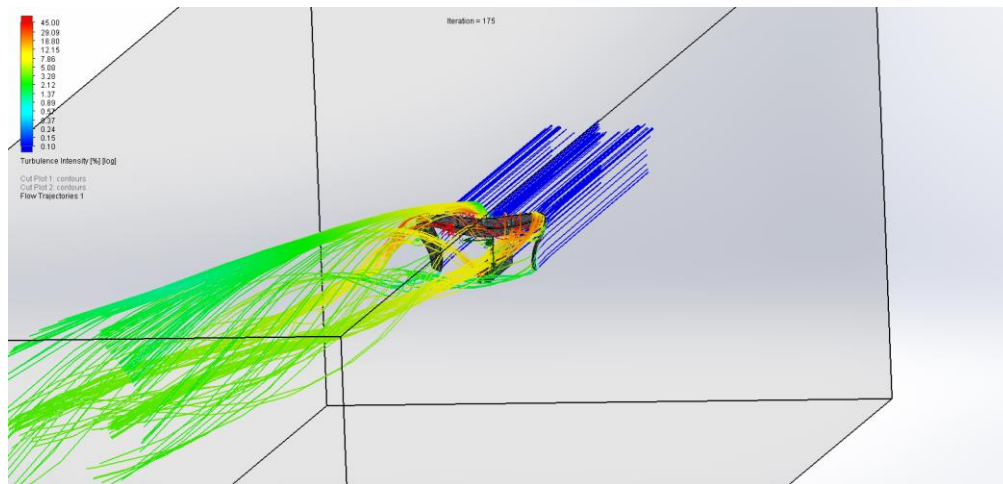


Figure 13: flow turbulence, back view

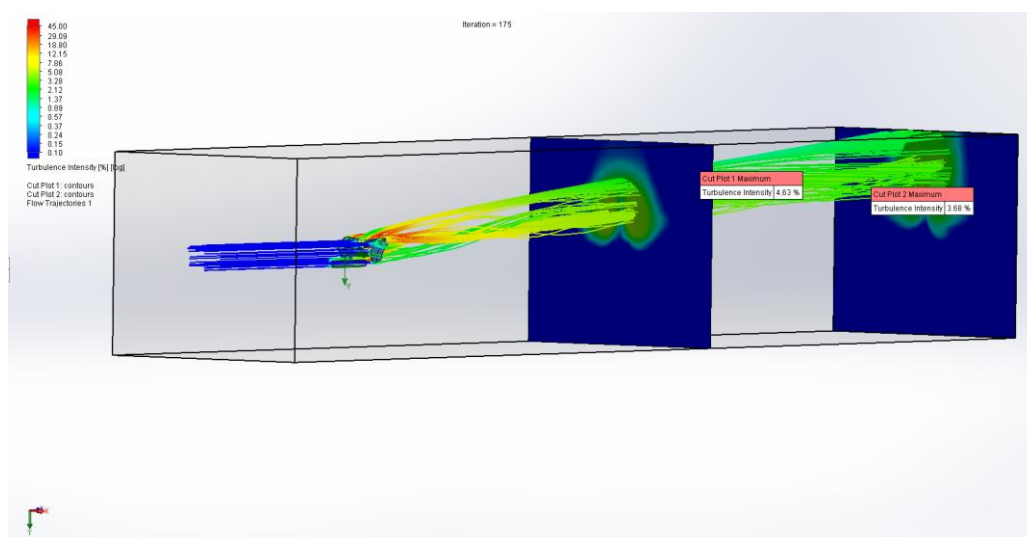


Figure 14: turbulence intensity.

As we can see in these images, the airflow experiences significant turbulence after passing through the rear wing. In fact, we can observe that, upon exiting, the turbulence is clearly present in the form of vortices. Their intensity exceeds 45% before decreasing to 4.63% at 10 meters and then to 3.68% at 20 meters. This ensures compliance with regulations, although we could further optimize the airflow since, at 4.63% at 10 meters, we are well below the permitted threshold of 18%.

- Forces:

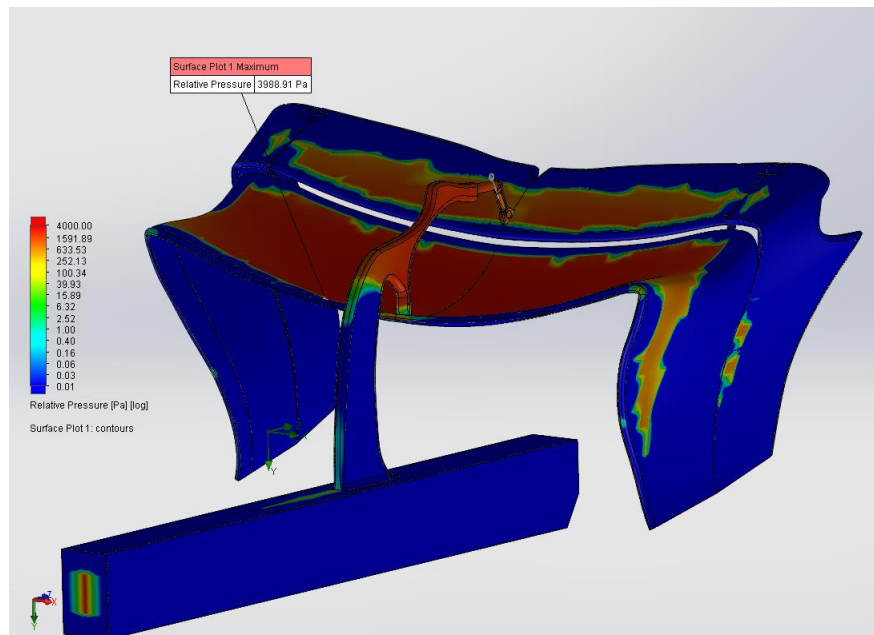


Figure 15: relative pressure on rear wing.

Summary									
Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Force (Y) 1	[N]	1656.109	1685.598	1637.092	1766.592	100	Yes	88.033	94.273
GG Force (Z) 2	[N]	777.457	796.024	776.401	818.477	100	Yes	17.802	69.226

Figure 16: forces on rear wing.

Based on the results obtained from the simulation, we have an aerodynamic downforce of 1656N and a drag force of 796N.

With DRS:

For our simulation under Monaco conditions, with opening DRS, we obtained the following results:

- Flow trajectorie:

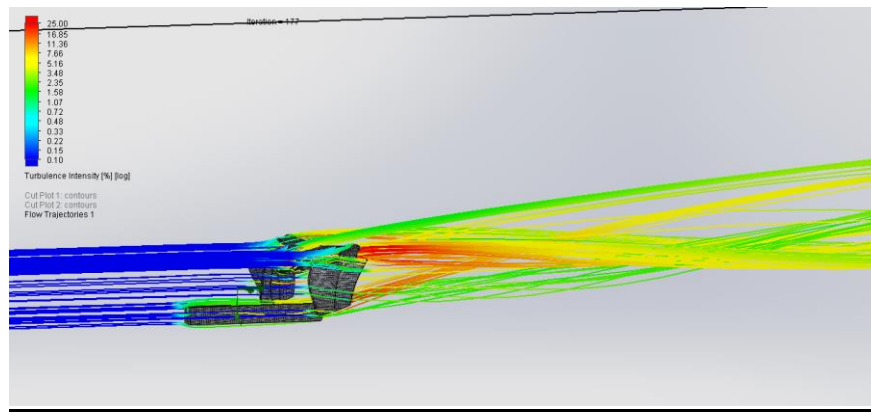


Figure 17: flow turbulence, front view.

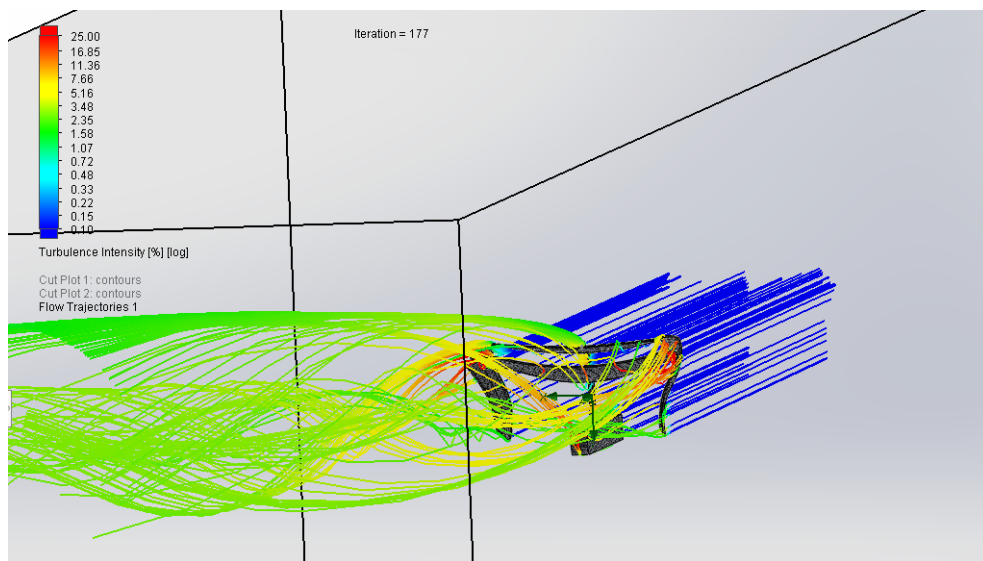


Figure 18: flow turbulence, back view.

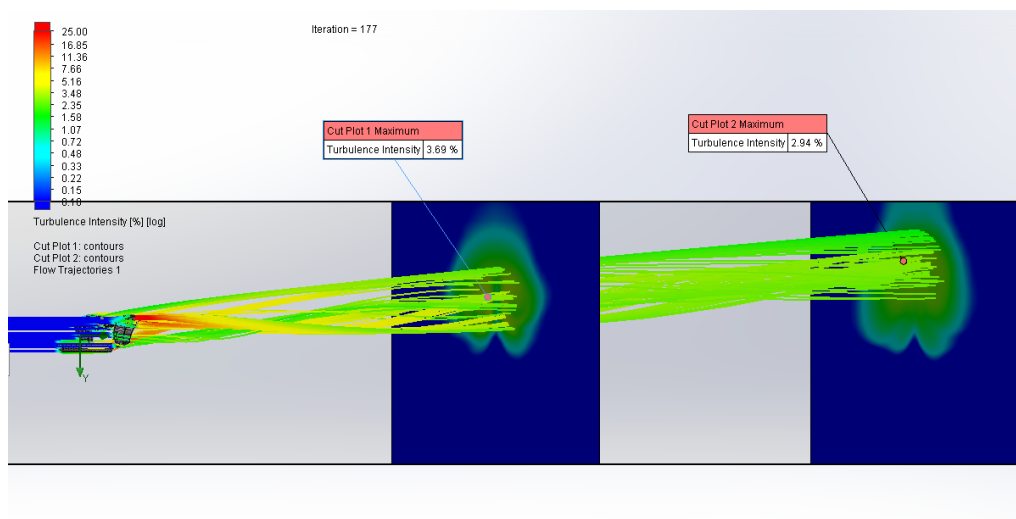


Figure 19: turbulence intensity.

As we can see in these images, the airflow experiences significant turbulence after passing through the rear wing but less than without DRS. In fact, we can

observe that, upon exiting, the turbulence is clearly present in the form of vortices. Their intensity exceeds 25% before decreasing to 3.69% at 10 meters and then to 2.94% at 20 meters. This ensures compliance with regulations, although we could further optimize the airflow since, at 3.69% at 10 meters, we are well below the permitted threshold of 18%.

- Forces:

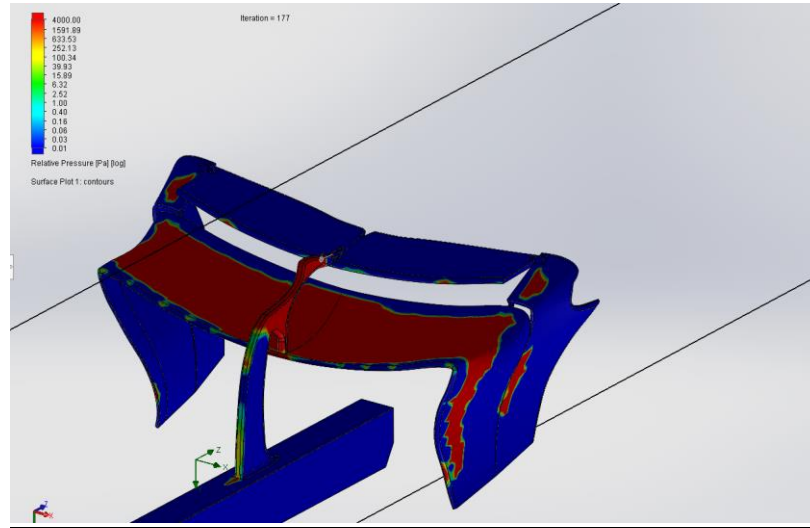


Figure 20: relative pressure on rear wing.

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Force (Y) 1	[N]	1329,02	1298,485	1175,595	1371,551	100	Yes	29,911	32,475
GG Force (Z) 2	[N]	510,742	514,206	504,273	534,027	100	Yes	83,539	21,691

Based on the results obtained from the simulation, we have an aerodynamic downforce of 1298N and a drag force of 514N.

Mexico :

Without DRS:

For our simulation under Mexico conditions, without DRS, we obtained the following results:

- Flow trajectorie:

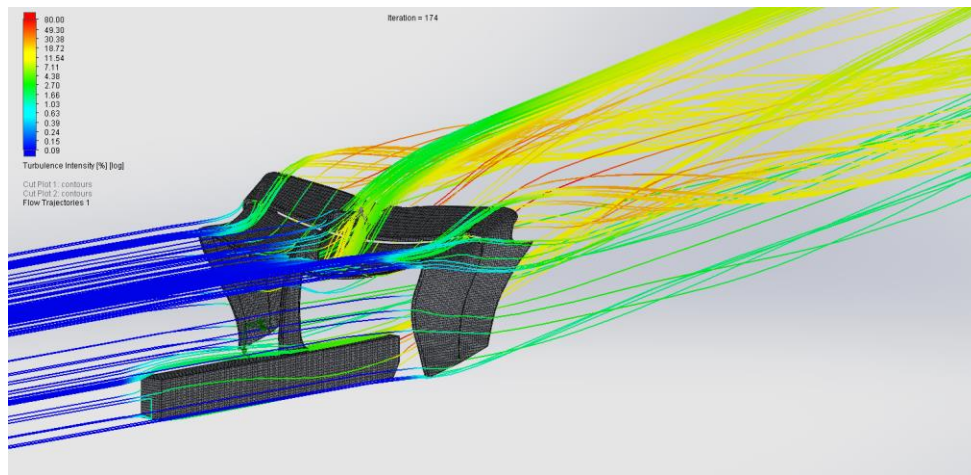


Figure 21: flow turbulence, front view.

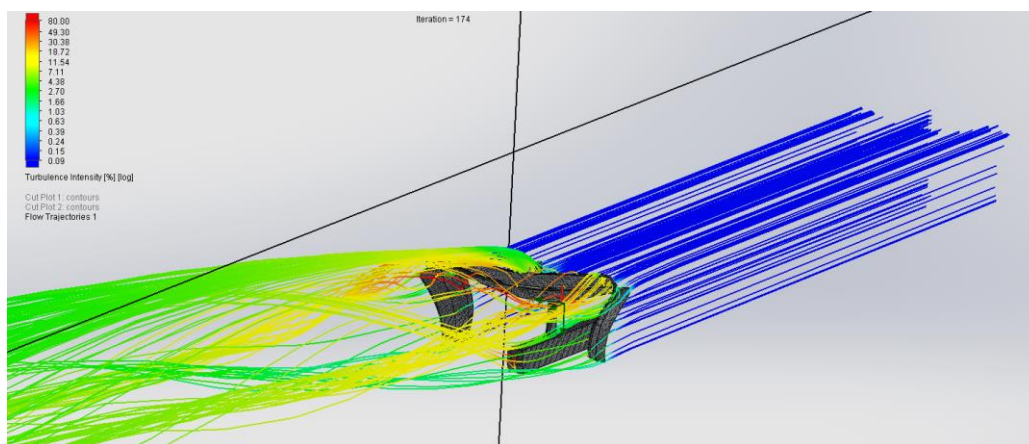


Figure 22: flow turbulence, back view.

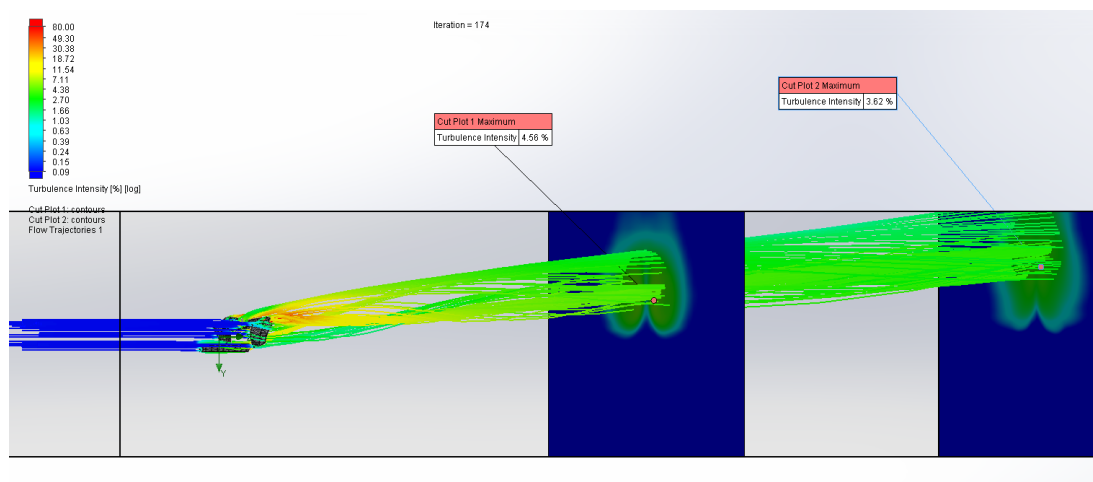


Figure 23: turbulence intensity.

As we can see in these images, the airflow experiences significant turbulence after passing through the rear wing but less than without DRS. In fact, we can observe that, upon exiting, the turbulence is clearly present in the form of

vortices. Their intensity exceeds 50% before decreasing to 4.56% at 10 meters and then to 3.62% at 20 meters. This ensures compliance with regulations, although we could further optimize the airflow since, at 4.56% at 10 meters, we are well below the permitted threshold of 18%.

- Forces:

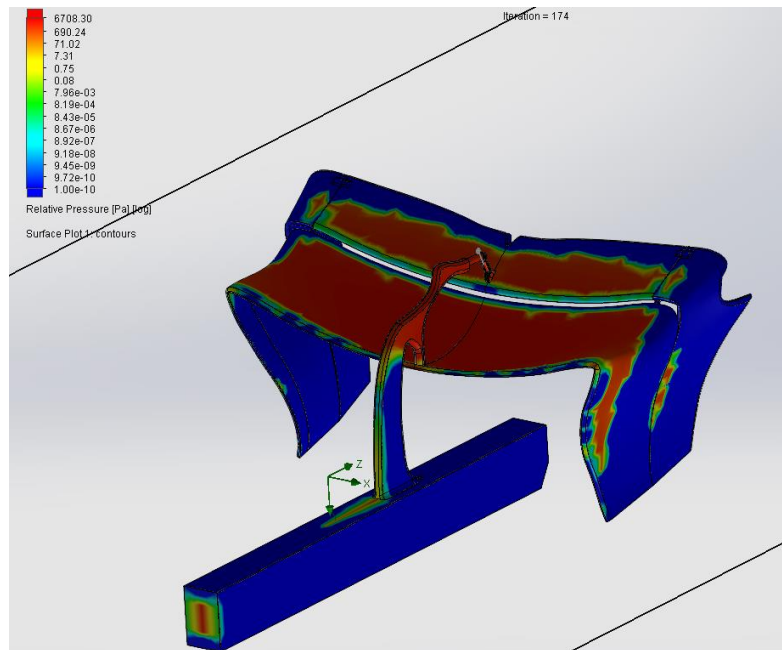


Figure 24: relative pressure on rear wing.

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Force (Y) 1	[N]	2176.491	2153.776	2077.043	2245.839	100	Yes	124.780	136.198
GG Force (Z) 2	[N]	1011.225	1018.321	995.746	1046.241	100	Yes	26.874	94.527

Based on the results obtained from the simulation, we have an aerodynamic downforce of 2154N and a drag force of 1018N.

With DRS:

For our simulation under Mexico conditions, with DRS, we obtained the following results:

- Flow trajectorie:

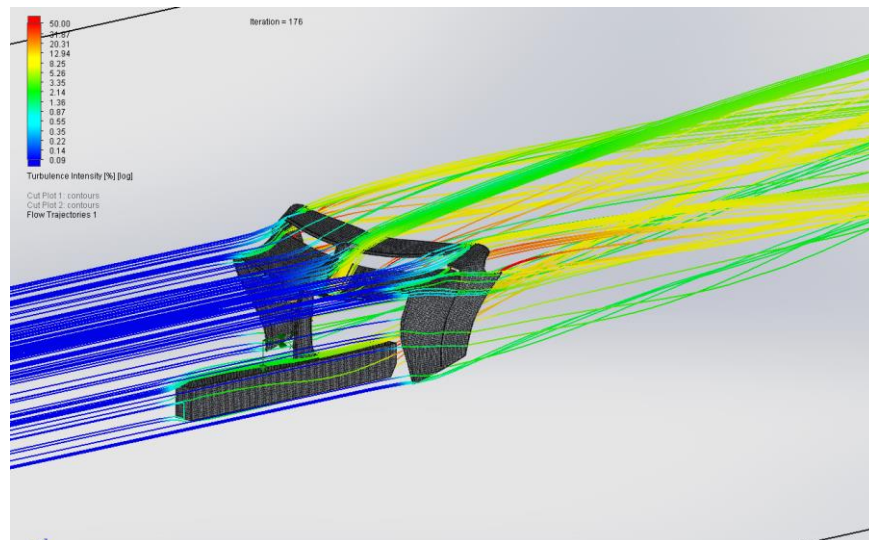


Figure 26: flow turbulence, front view.

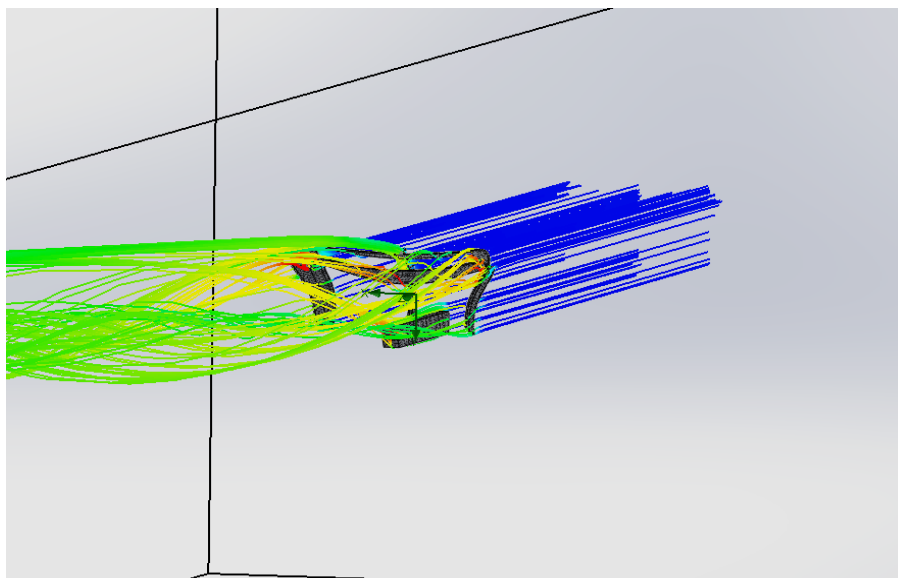


Figure 27: flow turbulence, back view.

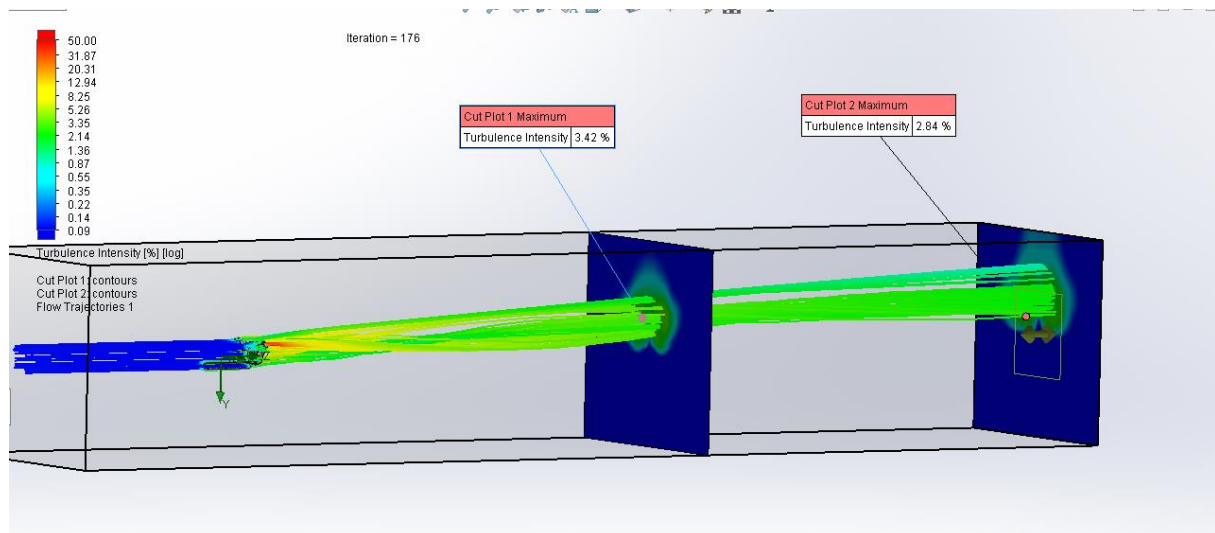


Figure 28: turbulence intensity.

As we can see in these images, the airflow experiences significant turbulence after passing through the rear wing but less than without DRS. In fact, we can observe that, upon exiting, the turbulence is clearly present in the form of vortices. Their intensity exceeds 50% before decreasing to 3.42% at 10 meters and then to 2.94% at 20 meters. This ensures compliance with regulations, although we could further optimize the airflow since, at 3.42% at 10 meters, we are well below the permitted threshold of 18%.

- Forces:

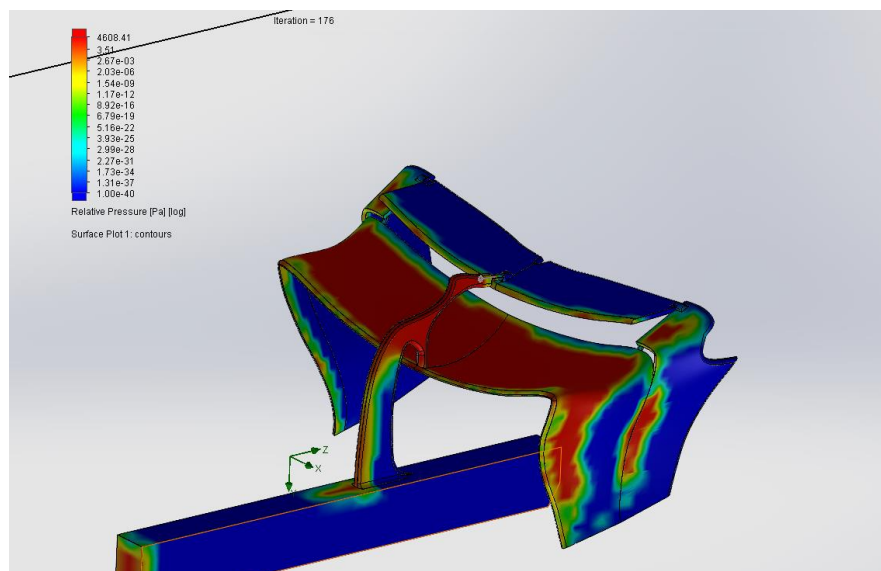


Figure 29: relative pressure on rear wing.

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Force (Y) 1	[N]	1654.576	1590.536	1516.146	1667.427	100	Yes	51.779	55.704
GG Force (Z) 2	[N]	679.055	669.995	653.142	681.203	100	Yes	9.008	35.309

Based on the results obtained from the simulation, we have an aerodynamic downforce of 1590N and a drag force of 670N.

2-Results and discussion:

Maintenant que nous avons tous les résultats de nos simulations, nous pouvons les analyser, les comparer et conclure de la véracité de notre model.

- Monaco :

	Turbulence (10m)	Turbulence (20m)	Down force (N)	Drag (N)
2021 no DRS	x	x	2550	850
2024 no DRS	4.63	3.68	1656	796
2024 DRS	3.69	2.94	1298	514

- Mexico :

	Turbulence (10m)	Turbulence (20m)	Down force (N)	Drag (N)
2021 no DRS	x	x	7170	2380
2024 no DRS	4.56	3.52	2154	1018
2024 DRS	3.42	2.94	1590	670

Monaco (2024)

1. Environmental Conditions:

- **Ambient Temperature:** 20.05°C
- **Atmospheric Pressure:** 1005 hPa
- **Air Density:** 1.225 kg/m³
- **Maximum Speed:** 289.9 km/h (80.528 m/s) by Mercedes

2. Turbulence:

- At **10m**, the turbulence is **4.63%** without DRS and reduces to **3.69%** with DRS.
- At **20m**, the turbulence decreases from **3.68%** to **2.94%** with DRS. These values indicate that the effect of DRS on the air flow is significant, especially in reducing the intensity of turbulence over distance.

3. Downforce:

- Without DRS, the **downforce** is **1656 N**, while it decreases to **1298 N** when DRS is activated. This reduction of **358 N** is expected, as the purpose of the DRS is to reduce downforce, thereby allowing for higher speeds in straights.

4. Drag:

- The **drag force** is **796 N** without DRS, which reduces to **514 N** when DRS is deployed, showing a decrease of **282 N**. The decrease in drag is consistent with the reduction of

the angle of the rear wing, optimizing the car's aerodynamics for less resistance and more speed on straights.

5. Summary for Monaco:

- The results show a strong impact of DRS on reducing both downforce and drag. The relatively high air density (1.225 kg/m^3) at sea level means the car faces significant aerodynamic forces, but the adjustments provided by DRS enable better performance in terms of speed while maintaining overall stability in the slower corners of Monaco.

Mexico (2024)

1. Environmental Conditions:

- **Ambient Temperature:** 20°C
- **Atmospheric Pressure:** 1027 hPa
- **Air Density:** 0.977 kg/m^3 (due to the 2300m altitude)
- **Maximum Speed:** 362.5 km/h (99 m/s) by McLaren

2. Turbulence:

- At **10m**, the turbulence is **4.56%** without DRS and drops to **3.42%** with DRS.
- At **20m**, the turbulence decreases from **3.52%** to **2.94%**. Although the turbulence reduction is somewhat less dramatic than in Monaco, it still confirms that DRS helps in smoothing the airflow at distance.

3. Downforce:

- Without DRS, the **downforce** is **2154 N**, while it drops to **1590 N** with DRS, a reduction of **564 N**. This is a more significant drop compared to Monaco, which can be attributed to the higher speeds and greater aerodynamic requirements on a faster circuit like Mexico.

4. Drag:

- The **drag force** is **1018 N** without DRS, which decreases to **670 N** with DRS, showing a reduction of **348 N**. The reduction is slightly more pronounced than in Monaco, reflecting the higher maximum speeds and the associated need for lower drag in a high-speed circuit like Mexico.

5. Summary for Mexico:

- At higher altitudes (2300 meters), the lower air density (0.977 kg/m^3) contributes to less air resistance overall, which could explain the higher maximum speed achieved by McLaren (362.5 km/h). The reduction in both downforce and drag due to DRS is also evident here, with a more significant decrease in both forces compared to Monaco, which supports the need for a more efficient aerodynamic setup to achieve higher speeds on the long straights in Mexico.

Key Discussion Points

1. Environmental Impact:

- The differences in **air density** between the two locations are crucial. In Monaco, the air density is **1.225 kg/m³** (at sea level), which provides more resistance and thus, more aerodynamic forces on the car. In contrast, Mexico has a lower air density of **0.977 kg/m³** due to its higher altitude (2300m), leading to a reduction in both drag and downforce, allowing for higher top speeds but also requiring greater reliance on aerodynamic efficiency to manage car stability and grip.

2. Turbulence:

- The effect of **DRS** is visible in both locations, reducing turbulence significantly at both 10m and 20m distances. This is especially critical for the drivers' ability to follow other cars closely and make overtakes on straights.

3. Downforce and Drag:

- **Monaco:** The **downforce** in Monaco is critical for stability through the tight corners. The reduction in downforce with DRS (358 N) helps the car achieve higher top speeds on the straights, but Monaco's narrow circuit means that excessive downforce reduction could negatively affect cornering stability.
- **Mexico:** The downforce is higher without DRS due to the need for greater stability at higher speeds. The larger reduction in downforce (564 N) when DRS is activated makes sense for a circuit with long straights, where minimizing drag is a priority.

4. Impact of DRS:

- In both cases, DRS helps reduce **drag** and **downforce**, improving straight-line speed. However, the larger reductions in downforce and drag observed in **Mexico** indicate that the aerodynamic benefits of DRS are more critical for circuits with long straights and higher speeds.
-

The differences observed between the 2021 and 2024 results can be attributed to several factors, including changes in the regulations, as well as the distinct simulation conditions used for each year.

First, the adjustments made to comply with the new aerodynamic regulations for 2024 naturally led to differences in downforce and drag values. These changes aim to reduce the amount of downforce generated by the vehicles and optimize aerodynamic efficiency in response to the new constraints imposed by the International Automobile Federation (FIA).

However, beyond the regulatory differences, the conditions under which the simulations were conducted play a significant role in the observed discrepancies. The 2021 results were obtained from laboratory tests conducted under controlled conditions, at a temperature of 15°C and atmospheric pressure of 1013 hPa, which may not reflect the actual race conditions. In fact, during the 2024 Monaco Grand Prix, the ambient temperature was 20.05°C and the atmospheric pressure was 1005 hPa, while in Mexico, the conditions were even more variable, with an altitude of 2300 meters, leading to a much lower air density (0.977 kg/m³) compared to the laboratory conditions of 2021. These variations in temperature and pressure directly influence air density, which can significantly impact the aerodynamic performance of the cars.

Thus, the differences observed between the 2021 and 2024 values are not only due to regulatory changes but also to differences in environmental conditions that directly affect the simulation results. It is therefore essential to consider these factors when analyzing the data, as they can explain part of the discrepancies in the car performances between different seasons.

Conclusion

In conclusion, the analysis of the aerodynamic performance of Formula 1 cars, particularly the study of drag and downforce, highlights the significant impact of the Drag Reduction System (DRS). The simulations conducted on the Monaco and Mexico circuits show that activating the DRS leads to a notable reduction in drag and downforce, allowing the cars to gain speed, especially in track sections where overtaking opportunities are most favorable.

However, it is important to note that the differences observed between the 2021 and 2024 seasons are also due to regulatory adjustments, which have influenced the aerodynamic configuration of the vehicles. The results obtained from these simulations, while adhering to the new standards, are also impacted by the specific environmental conditions at each circuit. Indeed, variations in temperature and pressure, as observed in Monaco and Mexico, directly affect air density and, consequently, the aerodynamic efficiency of the vehicles.

This emphasizes the importance of precise management of the DRS and environmental conditions during races. The DRS, by reducing drag and downforce, allows for higher top speeds and facilitates overtaking maneuvers, but it can also alter the car's aerodynamic balance in varied conditions. As such, this system remains a major strategic tool during races, and its influence must be considered holistically to maximize team performance throughout the season.

In summary, the effect of the DRS is undeniably beneficial in terms of performance, but its use must be carefully calibrated, taking into account new regulations and the specificities of each circuit to fully capitalize on its advantages.

1. **Clarke, D., & Smith, M. (2018).** *The Aerodynamics of Formula 1 Cars*. Springer.
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